HW-5 ASSIGNMENT-4

Q3 - $\int_0^{2\pi} \delta(a-b\cos\theta)\,d\theta$

$= \lim_{\epsilon\to 0} \frac{\epsilon}{\pi}\int_0^{2\pi} \frac{d\theta}{(b\cos\theta)^2 - 2ab\cos\theta + a^2+\epsilon^2}$

$z = e^{i\theta}$

$dz = i e^{i\theta} d\theta$

$d\theta = \frac{dz}{iz}$

$(b\cos\theta)^2 = \frac{b^2}{4}\left(z^2+\frac{1}{z^2}+2\right)$

$a - b\cos\theta = a - \frac{b}{2}\left(z+\frac{1}{z}\right) = \frac{2az - b(z^2+1)}{2z}$

$= \lim_{\epsilon\to 0} \frac{\epsilon}{\pi}\int_0^{2\pi} \frac{\cancel{d\theta} \times 4z^{\cancel{2}}}{(2az-b(z^2+1))^2+(2\epsilon z)^2} \times \frac{dz}{i\cancel{z}}$

$= \lim_{\epsilon\to 0} \frac{\epsilon}{i\pi}\int_0^{2\pi} \frac{4z\,dz}{(2az-b(z^2+1))^2+(2\epsilon z)^2}$

Denominator vanishes at $(2az-b(z^2+1)) \pm 2i\epsilon z = 0$

$bz^2 - 2(a\pm i\epsilon)z + b = 0$

$z_\pm^{(1,2)} = \frac{(a\pm i\epsilon) \pm \sqrt{(a\pm i\epsilon)^2 - b^2}}{b}$

As $\epsilon \to 0$

$z_\pm = \frac{a \pm \sqrt{a^2-b^2}}{b}$

$\cos\theta_0 = \frac{1}{2}\left(\frac{a\pm\sqrt{a^2-b^2}}{b} + \frac{b}{a\pm\sqrt{a^2-b^2}}\right)$

$= \frac{1}{2}\left(\frac{a^2 + a^2 \cancel{-b^2} \pm 2a\sqrt{a^2-b^2} \cancel{+b^2}}{b(a\pm\sqrt{a^2-b^2})}\right)$

$= \frac{\cancel{2}}{\cancel{2}} \times \frac{a\cancel{(a\pm\sqrt{a^2-b^2})}}{b\cancel{(a\pm\sqrt{a^2-b^2})}} = \frac{a}{b}$

$D(z)$ = Denominator

$I = \lim_{\epsilon\to 0}\frac{\epsilon}{\pi}\cdot\frac{2\pi \cancel{i}}{\cancel{i}}\sum \text{Res}\left(\frac{4z}{D(z)}, z_k\right)$

$D'(z_k) \approx i4\epsilon z_k \times 2(a-bz)$ $\cancel{= 4\epsilon}$

$I = \lim_{\epsilon\to 0}\frac{\epsilon}{\pi} \times \sum \frac{\cancel{2}}{i\cdot i 4\cancel{\epsilon} \times \cancel{2}(a-bz_k)}$

$$I = \frac{1}{b\sin\theta_0} + \frac{1}{b\sin\theta_0}$$

$$= \frac{2}{b\sin\theta_0}$$

$$= \frac{2}{b\sqrt{1-\frac{a^2}{b^2}}} = \cancel{2\sqrt{b^2}}$$

$$= \frac{2}{\sqrt{b^2-a^2}} \quad \text{for } b > a$$

For $b < a \Rightarrow \cos\theta = a/b$ has no soln.

so no pole on the contour

$\Rightarrow I = 0$

HW-7

Q1 - (ii)

a) $$\left[x^2\frac{d^2}{dx^2} + 4x\frac{d}{dx} + x^2 + 2\right] y(x) = 0$$

$$y(x) = \sum_{\lambda=0}^{\infty} a_\lambda x^{k+\lambda}$$

$$y' = \sum_{\lambda=0}^{\infty} a'_\lambda (k+\lambda)\, x^{k+\lambda-1} \qquad y'' = \sum_{\lambda=0}^{\infty} a_\lambda (k+\lambda)(k+\lambda-1)\, x^{k+\lambda-2}$$

$$\sum_{\lambda=0}^{\infty}\left[a_\lambda(k+\lambda)(k+\lambda-1)x^{k+\lambda} + 4a_\lambda(k+\lambda)x^{k+\lambda} + a_\lambda x^{k+\lambda+2} + 2a_\lambda x^{k+\lambda}\right] = 0$$

$\lambda = 0$:

$$a_0(k)(k-1) + 4a_0 k + \cancel{[illegible]}\ 2a_0 = 0$$

$$a_0(k^2+3k+2) = 0$$

$a_0 \neq 0$

$\Rightarrow k = -1$ or -2

$$x^{k+j}: \quad a_j(k+j)(k+j-1) \cancel{[illegible]} + 4a_j(k+j) + a_{j-2} + 2a_j = 0$$

$$a_j = \frac{-\,a_{j-2}}{(k+j)^2 + 3(k+j) + 2}$$

$k = -2$:

$$a_j = \frac{-a_{j-2}}{(j-2)^2+3(j-2)+2}$$

$$= \frac{-a_{j-2}}{j(j-1)}$$

$$a_{2m} = \frac{-a_{2m-2}}{2m(2m-1)} \Rightarrow a_{2m} = \frac{(-1)^m a_0}{(2m)!}$$

$$a_{2m+1} = \frac{(-1)^m a_1}{(2m+1)!}$$

$$y_1(x) = \sum_{\lambda=0}^{\infty} a_\lambda x^{k-2}$$

$$= x^{-2} \sum_{\lambda=0}^{\infty} a_{2m} x^{2m} + x^{-2} \sum_{\lambda=0}^{\infty} a_{2m+1} x^{2m+1}$$

$$= x^{-2}\left[a_0 \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m)!} x^{2m} + a_1 \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)!} x^{2m+1}\right]$$

$$= x^{-2}(a_0 \cos x + a_1 \sin x)$$

$k = -1$:

$$a_j = \frac{-a_{j-2}}{(j-1)^2+3(j-1)+2} = \frac{-a_{j-2}}{j^2-2j+\cancel{1}+3j\cancel{-1}}$$

$$= \frac{-a_{j-2}}{j(j+1)}$$

$$a_{2m} = \frac{-a_{2m-2}}{2m(2m+1)} \Rightarrow \cancel{a_{2m} = 0}$$

$$y_2(x) = x^{-2}(a_0 \sin x + a_1 \cos x)$$

$$y_1(x) = \frac{\sin x}{x^2}, \quad y_2(x) = \frac{\cos x}{x^2}$$

b) $$\left[x^2\frac{d^2}{dx^2} + 2x\frac{d}{dx} + x^2 - 2\right]y(x) = 0$$

$$y(x) = \sum_{\lambda=0}^{\infty} a_\lambda x^{k+\lambda}$$

$$\sum_{\lambda=0}^{\infty}\left[a_\lambda (k+\lambda)(k+\lambda-1)x^{k+\lambda} + 2a_\lambda(k+\lambda)x^{k+\lambda} + a_\lambda x^{k+\lambda+2} - 2a_\lambda x^{k+\lambda}\right] = 0$$

$\lambda = 0$:

$$\cancel{a_0}(k)(k-1) + 2\cancel{a_0}k - 2\cancel{a_0} = 0$$

$a_0 \neq 0$

$$k^2 - k + 2k - 2 = 0$$

$$k^2 + k - 2 = 0$$

$$(k+2)(k-1) = 0$$

$$k = 1 \text{ or } -2$$

a^{k+j} : $$a_j(k+j)(k+j-1) + 2a_j(k+j) + a_{j-2} - 2a_j = 0$$

$$a_j = \frac{-a_{j-2}}{(k+j)^2 + (k+j) - 2}$$

$k = 1$:

$$a_j = \frac{-a_{j-2}}{j(j+3)}$$

Odd terms go to zero

$$y_1(x) = a_0\left(x^2 - \frac{x^4}{30} + \frac{x^6}{840} - \cdots\right)$$

$k = -2$: $$a_j = -\frac{a_{j-2}}{j(j-3)}$$

Odd terms must be zero

$$y_2(x) = a_0\left(\frac{x^2}{2} - \frac{x^4}{8} + \frac{x^6}{144} - \cdots\right)$$

(iv) $\left[\frac{d^2}{dx^2} - \frac{m(m+1)}{x^2}\right] y(x) = 0$

$$y(x) = \sum_{\lambda=0}^{\infty} a_\lambda x^{k+\lambda}$$

$$\sum_{\lambda=0}^{\infty} \left[a_\lambda (k+\lambda)(k+\lambda-1) x^{k+\lambda-2} - m(m+1) a_\lambda x^{k+\lambda-2} \right] = 0$$

x^{k-2} : $a_0 k(k-1) - m(m+1) a_0 = 0$

$$k^2 - k - m(m+1) = 0$$

$$k = \frac{1 \pm \sqrt{1+4m(m+1)}}{2} = \frac{1 \pm (2m+1)}{2}$$

$k = m+1$ or $-m$

x^{k+j} : $a_{j+2}(k+j+2)(k+j+1) - m(m+1) a_{j+2} = 0$

$k = m+1$:

$$a_{j+2}\left[(m+3+j)(m+j+2) - m(m+1)\right] = 0$$

$$a_{j+2}\left[\cancel{m^2} + mj + 2m + 3m + 3j + 6 + mj + j^2 + 2j - \cancel{m^2} - m\right] = 0$$

$$a_{j+2}\left[j^2 + j(2m+5) + (4m+6)\right] = 0$$

$\Rightarrow a_{j+2} = 0 \quad \forall \; j \geq -1$

$$y_1(x) = a_0 x^{m+1}$$

$k = -m$:

$$a_{j+2}\left[(m+j+2)(m+j+1) - m(m+1)\right] = 0$$

$\Rightarrow a_{j+2} = 0 \; \forall \; j \geq 1$

$$y_2(x) = b_0 x^{-m}$$

Q2 - $J_m(x) = \sum_{j=0}^{\infty} \frac{(-1)^j}{j!\,(m+j)!} \left(\frac{x}{2}\right)^{m+2j}$

$$J_1(x) = \frac{x}{2} - \frac{1}{2}\times\frac{x^3}{8} + \frac{1}{2\times 6}\times\frac{x^5}{32} - \cdots$$

$$= \frac{x}{2} - \frac{x^3}{16} + \frac{x^5}{384} - \cdots$$

$$y_2(x) = y_1(x) \int_0^x \frac{e^{-\int^{x_2} P(x_1)\,dx_1}}{y_1^2(x_2)}\,dx_2 \qquad P(x_1) = \frac{1}{x_1}$$

$$e^{\int P(x)\,dx_1} = e^{-\ln x_2} = \frac{1}{x_2}$$

$$\int_0^x \frac{1}{x_2} \cdot \frac{dx_2}{\left(\frac{x_2}{2} - \frac{x_2^3}{8} + \frac{x_2^5}{384}\right)^2} = \int_0^x \frac{dx_2}{\frac{x_2^3}{4}\left(1 - \frac{x_2^2}{8} + \frac{x_2^4}{192}\right)^2}$$

$$f(x) = 1 - \frac{x_2^2}{8} + \frac{x_2^4}{192} = 1 + u(x)$$

$$(1+u)^{-2} = 1 + {}^{-2}C_1\,u + {}^{-2}C_2\,u^2 + {}^{-2}C_3\,u^3 + \cdots$$

$$= 1 - 2u + \frac{(\cancel{-2})(-3)}{\cancel{2!}}\,u^2 + \frac{(\cancel{-2})(\cancel{-3})(-4)}{\cancel{3!}}\,u^3$$

$$= 1 - 2u + 3u^2 - 4u^3$$

$$\frac{1}{f(x)^2} = 1 - 2\left(\frac{-x_2^2}{8} + \frac{x_2^4}{192}\right) + 3\left(\frac{-x_2^2}{8} + \frac{x_2^4}{192}\right)^2$$

$$= 1 + \frac{x_2^2}{4} - \frac{x_2^4}{96} + \frac{3x_2^4}{64} + \cdots$$

$$= 1 + \frac{x_2^2}{4} + \frac{7x_2^4}{192}$$

$$y_1(x) \int_0^x \frac{4}{x_2^3}\left(1 + \frac{x_2^2}{4} + \frac{7}{192}x_2^4\right) dx_2$$

$$= y_1(x) \int_0^x \left(\frac{4}{x_2^3} + \frac{1}{x_2} + \frac{7}{43}x_2\right) dx_2$$

$$= \left(\frac{x}{2} - \frac{x^3}{16} + \frac{x^5}{384}\right)\left(\frac{-2}{x^2} + \ln x + \frac{7}{96}x^2\right)$$

$$= -\frac{1}{x} + \frac{x}{2}\ln x + \frac{7}{192}x^3 + \frac{x}{8} - \frac{x^3}{16}\ln x - \frac{x^3}{192}$$

$$N_1(x) = -\frac{1}{x} + \frac{x}{8} + \frac{x}{2}\ln x + \frac{x^3}{32} - \frac{x^3}{16}\ln x$$

$$J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$$

$$N_{\frac{1}{2}}(x) = J_{\frac{1}{2}}(x) \int_0^x \frac{1}{x_2} \cdot \frac{dx_2}{\frac{2}{\pi \cancel{x_2}} \sin^2 x_2}$$

$$= \frac{\pi}{2} J_{\frac{1}{2}}(x) \int_0^x \frac{dx_2}{\sin^2 x_2}$$

$$= -\frac{\pi}{2} J_{\frac{1}{2}}(x) \times \frac{\cos x_2}{\sin x_2}$$

$$= -\frac{\pi}{2} \sqrt{\frac{2}{\pi x}} \cos x$$

$$= -\sqrt{\frac{\pi}{2}} \frac{\cos x}{\sqrt{x}}$$

Q3- $\frac{-\hbar^2}{2m} \nabla^2 \psi(\rho, \phi, z) = E \psi(\rho, \phi, z)$

$$\psi(\rho, \phi, z) = R(\rho) \Phi(\phi) Z(z)$$

$$\left[\frac{1}{R(\rho)\rho} \frac{d}{d\rho}\left(\rho \frac{dR}{d\rho}\right) + \frac{1}{\Phi(\phi)\rho^2} \frac{d^2\Phi}{d\phi^2} + \frac{1}{Z(z)} \frac{d^2 Z}{dz^2}\right] = \underbrace{-\frac{2mE}{\hbar^2}}_{k^2}$$

$$\frac{1}{R(\rho)\rho} \frac{d}{d\rho}\left(\rho \frac{dR}{d\rho}\right) + \frac{1}{\Phi(\phi)\rho^2} \frac{d^2\Phi}{d\phi^2} = -k^2 - \frac{1}{Z(z)} \frac{d^2 Z}{dz^2} = 0$$

Both indefendtly must be zero

$$\frac{d^2 Z}{dz^2} + k_z^2 Z = 0 \qquad (Z(0) = Z(H) = 0)$$

$$\Rightarrow \quad Z(z) = \sin(k_z z)$$

$$k_z = \frac{m\pi}{H}$$

Let $k_\perp^2 \equiv k^2 - k_z^2$

$$\rho \frac{1}{R(\rho)} \frac{d}{d\rho}\left(\rho \frac{dR}{d\rho}\right) + \frac{1}{\Phi(\phi)} \frac{d^2\Phi}{d\phi^2} + k_\perp^2 \rho^2 = 0$$

$$\frac{1}{\Phi(\phi)} \frac{d^2\Phi}{d\phi^2} = -\left[\rho \frac{1}{R} \frac{d}{d\rho}\left(\rho \frac{dR}{d\rho}\right) + k_\perp^2 \rho^2\right] = -m^2$$

$$\frac{d^2\Phi}{d\phi^2} + m^2 \Phi = 0$$

$$\Phi(\phi) = e^{im\phi}$$

$$\frac{\rho}{R(\rho)}\left(\frac{dR}{d\rho}+\rho\frac{d^2R}{d\rho^2}+k_\perp^2\rho\right)=m^2$$

$$\rho^2R''+\rho R'+(k_\perp^2\rho^2-m^2)R=0$$

$$R''+\frac{1}{\rho}R'+\left(k_\perp^2-\frac{m^2}{\rho^2}\right)R=0$$

$$R(\rho)\propto J_m(k_\perp\rho)$$

$R(R)=0 \longrightarrow$ zeros of J_m

$k_\perp = \dfrac{\alpha_{m,s}}{R}$; $\alpha_{m,s} \longrightarrow s^{th}$ root of J_m

$$E_{m,s,n}=\frac{\hbar^2}{2m}\left(k_\perp^2+k_z^2\right)$$

$$=\frac{\hbar^2}{2m}\left(\frac{\alpha_{m,s}^2}{R^2}+\frac{n^2\pi^2}{H^2}\right)$$

Q4- $$e^{ix\cos\phi}=\sum_{n=-\infty}^{\infty}i^n J_n(x)\,e^{in\phi}$$

$$J_{-n}(x)=(-1)^n J_n(x)$$

$$\cos(x\cos\phi)=\mathrm{Re}\left(e^{ix\cos\phi}\right)$$

$$=\sum_{n=-\infty}^{\infty}J_n(x)\,\mathrm{Re}\left(i^n e^{in\phi}\right)$$

$$=\sum_{n=-\infty}^{\infty}J_n(x)\,\mathrm{Re}\left(e^{in(\phi+\frac{\pi}{2})}\right)$$

$$=\sum_{n=-\infty}^{\infty}J_n(x)\cos\left(n\phi+\frac{n\pi}{2}\right)$$

$$=J_0(x)+\sum_{n=1}^{\infty}J_n(x)\cos\left(n\phi+\frac{n\pi}{2}\right)+\sum_{n=-1}^{\infty}J_{-n}(x)\cos\left(n\phi+\frac{n\pi}{2}\right)$$

$$=J_0(x)+2\sum_{n=1}^{\infty}J_{2n}(x)\cos\left(2n\phi+n\pi\right)$$

$$=J_0(x)+2\sum_{n=1}^{\infty}(-1)^n J_{2n}(x)\cos(2n\phi)$$

$$\sin(x\cos\phi) = \sum_{m=-\infty}^{\infty} J_m(x)\sin\left(m\phi + \frac{m\pi}{2}\right)$$

$$= \sum_{m=0}^{\infty} J_m(x)\sin\left(m\phi + \frac{m\pi}{2}\right) - (-1)^m J_m(x)\sin\left(m\phi + \frac{m\pi}{2}\right)$$

Only odd terms survive:

$$= 2\sum_{m=1}^{\infty} J_{2m-1}\sin\left((2m-1)\phi + (2m-1)\frac{\pi}{2}\right)$$

$$= 2\sum_{m=1}^{\infty} (-1)^{m+1} J_{2m-1}(x)\cos[(2m-1)\phi]$$

Q6- $\left[\nabla_r^2 + k^2\right]\psi(r,\theta,\phi) = 0$

$k^2 = \frac{2ME}{\hbar^2}$; $\psi(r,\theta,\phi) = R(r)P(\theta)\Phi(\phi)$

a) $$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)P\Phi + \frac{1}{r^2\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{dP}{d\theta}\right)R\Phi + \frac{1}{r^2\sin^2\theta}\frac{d^2\Phi}{d\phi^2}RP + k^2RP\Phi = 0$$

Dividing by $RP\Phi$ & Multiplying with r^2 $\Rightarrow$

$$\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + k^2r^2 + \frac{1}{P\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{dP}{d\theta}\right) + \frac{1}{\Phi\sin^2\theta}\frac{d^2\Phi}{d\phi^2} = 0$$

$$\Rightarrow \frac{1}{R(r)}\frac{d}{dr}\left(r^2\frac{dR(r)}{dr}\right) + k^2r^2 = -\frac{1}{P(\theta)\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{dP(\theta)}{d\theta}\right) + \frac{1}{\Phi(\phi)\sin^2\theta}\frac{d^2\Phi(\phi)}{d\phi^2} = Q$$

$$\frac{1}{P(\theta)\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{dP}{d\theta}\right) + \frac{1}{\Phi(\phi)\sin^2\theta}\frac{d^2\Phi}{d\phi^2} + Q = 0$$

Multiplying by $\sin^2\theta$

$$\sin\theta\frac{1}{P(\theta)}\frac{d}{d\theta}\left(\sin\theta\frac{dP}{d\theta}\right) + Q\sin^2\theta + \frac{1}{\Phi(\phi)}\frac{d^2\Phi}{d\phi^2} = 0$$

$$\frac{1}{\Phi(\phi)}\frac{d^2\Phi}{d\phi^2} = -\left[\frac{\sin\theta}{P(\theta)}\frac{d}{d\theta}\left(\sin\theta\frac{dP}{d\theta}\right) + Q\sin^2\theta\right] = -m^2$$

$$\Rightarrow \frac{1}{\Phi(\phi)}\frac{d^2\Phi}{d\phi^2} = -m^2$$

$$\frac{d^2\Phi}{d\phi^2} + m^2\Phi = 0 \Rightarrow \boxed{\left[\frac{d^2}{d\phi^2} + m^2\right]\Phi(\phi) = 0}$$

$$\frac{1}{P(\theta)} \sin\theta \frac{d}{d\theta}\left(\sin\theta \frac{dP}{d\theta}\right) + Q \sin^2\theta - m^2 = 0$$

Multiplying by $P(\theta)$ & dividing by $\sin^2\theta$

$$\frac{1}{\sin\theta} \frac{d}{d\theta}\left(\sin\theta \frac{dP}{d\theta}\right) + Q\, P(\theta) - \frac{m^2}{\sin^2\theta} P(\theta) = 0$$

$$\boxed{\left[\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta \frac{d}{d\theta}\right) - \frac{m^2}{\sin^2\theta} + Q\right] P(\theta) = 0}$$

$$\frac{1}{R(r)} \frac{d}{dr}\left(r^2 \frac{dR}{dr}\right) + k^2 r^2 - Q = 0$$

Multiplying with $R(r)$

$$\frac{d}{dr}\left(r^2 \frac{dR}{dr}\right) + k^2 r^2 R(r) - Q R(r) = 0$$

$$\boxed{\left[\frac{d}{dr}\left(r^2 \frac{d}{dr}\right) + k^2 r^2 - Q\right] R(r) = 0}$$

b) We choose m^2 because:

1. The azimuthal eq^n. is second order so $+m$ & $-m$ give valide, linearly independent solns. $\Rightarrow$ const. must be symmetric in m
2. Guarantees real & oscillatory $e^{\pm im\phi}$ solns.
3. Allows integer m values corresponding to single-value wavefx^{n}s.

c) $R(x) = \frac{Z(x)}{\sqrt{x}}$; $x = kr$; $Q = m(m+1)$

$$R(r) = R\left(\frac{x}{k}\right) = \frac{Z(x)}{\sqrt{x}}$$

$$\frac{dR}{dx} = k \frac{d}{dx}\left(\frac{Z(x)}{\sqrt{x}}\right) = k\left(\frac{Z'(x)}{\sqrt{x}} - \frac{1}{2} Z(x)\, x^{-3/2}\right)$$

$$r^2 \frac{dR}{dx} = \cancel{k}\, \frac{x^2}{k^2} \cdot k\left(\frac{Z'(x)}{\sqrt{x}} - \frac{1}{2} Z\, x^{-3/2}\right) = \frac{1}{k}\left(x^{\frac{3}{2}} Z' - \frac{1}{2} x^{\frac{1}{2}} Z\right)$$

$$\frac{d}{dr}\left(r^2 \frac{dR}{dr}\right) = \frac{\cancel{k}}{\cancel{k}} \frac{d}{dx}\left(x^{\frac{3}{2}} Z' - \frac{1}{2} x^{\frac{1}{2}} Z\right)$$

$$= x^{\frac{3}{2}} Z'' + \frac{3}{2} x^{\frac{1}{2}} Z' - \frac{1}{4} x^{-1/2} Z - \frac{1}{2} x^{\frac{1}{2}} Z'$$

$$x^{\frac{3}{2}} Z'' + x^{\frac{1}{2}} Z' - \frac{1}{4} x^{-\frac{1}{2}} Z + \left[x^2 - n(n+1)\right] \frac{Z(x)}{\sqrt{x}} = 0$$

$$x^2 Z'' + x Z' - \frac{1}{4} Z + \left(x^2 - n(n+1)\right) Z = 0$$

$$x^2 Z'' + x Z' + \left(x^2 - \left(n + \frac{1}{2}\right)^2\right) Z = 0$$

$$\Rightarrow \left[x^2 \frac{d^2}{dx^2} + x \frac{d}{dx} + x^2 - \left(n + 1/2\right)^2\right] Z(x) = 0$$

$$j_0(x) = \frac{\sin x}{\sqrt{x}} \qquad P(x) = \frac{1}{x}$$

$$n_0(x) = j_0(x) \int_0^x \frac{e^{-\int_0^{x_2} P(x_1)\,dx_1}}{j_0^2(x_2)} \, dx_2$$

$$= j_0(x) \int_0^x \frac{1}{\cancel{x_2} \cdot \frac{\sin^2 x_2}{\cancel{x_2}}} \, dx_2$$

$$= j_0(x) \int_0^x \operatorname{cosec}^2 x_2 \, dx_2$$

$$= \frac{\sin x}{\sqrt{x}} \times -\cot x \quad = -\frac{\cancel{\sin x}}{\sqrt{x}} \times \frac{\cos x}{\cancel{\sin x}}$$

$$n_0(x) = -\frac{\cos x}{\sqrt{x}}$$

Q7-

$$j_n(x) = \sqrt{\frac{\pi}{2x}} \, J_{n+\frac{1}{2}}(x)$$

a) $\nu = n + \frac{1}{2}$

$$j_{n-1}(x) + j_{n+1}(x) = \sqrt{\frac{\pi}{2x}} \left[J_{n-\frac{1}{2}}(x) + J_{n+\frac{3}{2}}(x)\right]$$

$$= \sqrt{\frac{\pi}{2x}} \left[J_{\nu-1}(x) + J_{\nu+1}(x)\right]$$

$$= \sqrt{\frac{\pi}{2x}} \times \frac{2\nu}{x} J_\nu(x)$$

$$= \sqrt{\frac{\pi}{2x}} \times \frac{2\left(n + \frac{1}{2}\right)}{x} J_{n+\frac{1}{2}}(x)$$

$$j_{n-1}(x) + j_{n+1}(x) = \frac{2n+1}{x} j_n(x)$$

b) $$j_n'(x) = \sqrt{\frac{\pi}{2}}\left[-\frac{1}{2} x^{-\frac{3}{2}} J_\nu(x) + \frac{1}{\sqrt{x}} J_\nu'(x)\right]$$

$$= \sqrt{\frac{\pi}{2x}}\left(J_\nu'(x) - \frac{1}{2x}J_\nu(x)\right)$$

$$= \sqrt{\frac{\pi}{2x}}\left[\frac{J_{\nu-1}(x)-J_{\nu+1}(x)}{2} - \frac{1}{2\cancel{x}}\times\frac{\cancel{x}}{2\nu}\left(J_{\nu-1}(x)+J_{\nu+1}(x)\right)\right]$$

$$= \sqrt{\frac{\pi}{2x}}\left[\left(\frac{1}{2} - \frac{1}{4\nu}\right)J_{\nu-1}(x) + \left(-\frac{1}{2} - \frac{1}{4\nu}\right)J_{\nu+1}(x)\right]$$

$$= \sqrt{\frac{\pi}{2x}}\left[\left(\frac{2\nu-1}{4\nu}\right)J_{\nu-1}(x) - \left(\frac{2\nu+1}{4\nu}\right)J_{\nu+1}(x)\right]$$

Replacing $\nu \to n+\frac{1}{2}$

$$j_n'(x) = \sqrt{\frac{\pi}{2x}}\left[\frac{n}{2n+1}J_{n-\frac{1}{2}}(x) - \left(\frac{n+1}{2n+1}\right)J_{n+\frac{3}{2}}(x)\right]$$

$$j_n'(x) = \frac{n}{2n+1}j_{n-1}(x) - \frac{n+1}{2n+1}j_{n+1}(x)$$

c) $$(2n+1)\,j_n'(x) = n\,j_{n-1} - n\,j_{n+1} - j_{n+1}$$

$$j_{n-1} + j_{n+1} = \frac{2n+1}{x}j_n$$

$$j_{n-1} = \frac{2n+1}{x}j_n - j_{n+1}$$

$$(2n+1)j_n' = n\left(\frac{2n+1}{x}j_n - \cancel{j_{n+1}}\right) - \cancel{n\,j_{n+1}} - j_{n+1}$$

$$\cancel{(2n+1)}j_n' = \frac{n\cancel{(2n+1)}}{x}j_n - \cancel{(2n+1)}j_{n+1}$$

$$j_n' = \frac{n}{x}j_n - j_{n+1}$$

$$\Rightarrow \quad j_{n+1}(x) = \frac{n}{x}j_n(x) - j_n'(x)$$

$$j_{n-1}(x) = \frac{2n+1}{x}j_n - \frac{n}{x}j_n + j_n' = \frac{n+1}{x}j_n - j_n'$$

$$\therefore \quad j_{n\pm1}(x) = \left(n+\frac{1}{2}\mp\frac{1}{2}\right)\frac{1}{x}j_n(x) \mp j_n'(x)$$

d) $$\frac{d}{dx}\left[x^{m+1} j_m(x)\right] = (m+1)x^m j_m(x) + x^{m+1} j_m'(x)$$

$$= (m+1)x^m j_m(x) + x^{m+1}\left(j_{m-1}(x) - \frac{m+1}{x} j_m(x)\right)$$

$$= \cancel{(m+1)x^m j_m(x)} + x^{m+1} j_{m-1}(x) - \cancel{x^m (m+1) j_m(x)}$$

$$= x^{m+1} j_{m-1}(x)$$

e) $$\frac{d}{dx}\left[x^{-m} j_m(x)\right] = -m x^{-m-1} j_m(x) + x^{-m} j_m'(x)$$

$$= -m x^{-m-1} j_m(x) + x^{-m}\left[\frac{m}{x} j_m(x) - j_{m+1}(x)\right]$$

$$= \cancel{-m x^{-m-1} j_m(x)} + \cancel{m x^{-m-1} j_m(x)} - x^{-m} j_{m+1}(x)$$

$$= -x^{-m} j_{m+1}(x)$$

Q8- a) $$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta \frac{dP}{d\theta}\right) - \frac{m^2}{\sin^2\theta} P(\theta) + n(n+1)P(\theta) = 0$$

$x = \cos\theta$, $dx = -\sin\theta\, d\theta$

$$\frac{d}{d\theta} = \frac{dx}{d\theta}\frac{d}{dx} = -\sin\theta \frac{d}{dx}$$

$$\frac{dP}{d\theta} = -\sin\theta \frac{dP}{dx}$$

$$\sin\theta \frac{dP}{dx} = -\sin^2\theta \frac{dP}{dx} = -(1-x^2)\frac{dP}{dx}$$

$$\frac{d}{d\theta}\left(\sin\theta \frac{dP}{dx}\right) = \frac{d}{d\theta}\left(-(1-x^2)\frac{dP}{dx}\right) = -\left(\frac{dx}{d\theta}\right)\frac{d}{dx}\left[(1-x^2)\frac{dP}{dx}\right]$$

$$= \sin\theta \frac{d}{dx}\left[(1-x^2)\frac{dP}{dx}\right]$$

$$\frac{d}{dx}\left[(1-x^2)\frac{dP}{dx}\right] - \frac{m^2}{1-x^2}P(x) + n(n+1)P(x) = 0$$

$$(1-x^2)\frac{d^2P}{dx^2} - 2x\frac{dP}{dx} + \left(n(n+1) - \frac{m^2}{1-x^2}\right)P(x) = 0$$

$$\Rightarrow \left[(1-x^2)\frac{d^2}{dx^2} - 2x\frac{d}{dx} - \frac{m^2}{1-x^2} + n(n+1)\right]y(x) = 0$$

b) $y(x) = \sum_{j=0}^{\infty} a_j x^{j+k}$

$$(1-x^2)y'' \cancel{+} - 2xy' + m(m+1)y = 0$$

$$\sum_{j=0}^{\infty} (j+k)(j+k-1)a_j x^{k+j-2} + (j+k)(j+k-1)a_j x^{k+j} - 2(j+k)(j+k-1)x^{k+j} + m(m+1)a_j x^{k+j} = 0$$

$$(j+k+2)(j+k+1)a_{j+2} + \left[-(j+k)(j+k+1) + m(m+1)\right]a_j = 0$$

$$a_{j+2} = \frac{(j+k)(j+k+1) - m(m+1)}{(j+k+1)(j+k+2)}$$

c) $k = 0,\ a_1 = 0$

$\Rightarrow$ Odd terms vanish

$$a_{j+2} = \frac{j(j+1) - m(m+1)}{(j+1)(j+2)} a_j$$

$j=0:$ $a_2 = \frac{0 - m(m+1)}{1\cdot 2} a_0 = -\frac{m(m+1)}{2!} a_0$

$j=2:$ $a_4 = \frac{2\cdot 3 - m(m+1)}{3\cdot 4} a_2 = \frac{6 - m(m+1)}{12} a_2 = \frac{m(m+1)(m-2)(m+3)}{24} a_0$

$$= \frac{(m-2)m(m+1)(m+3)}{4!} a_0$$

$$y_1(x) = a_0\left[1 - \frac{m(m+1)}{2!}x^2 + \frac{(m-2)m(m+1)(m+3)}{4!}x^4 \dots\right]$$

$k = 1,\ a_1 = 0$

$\Rightarrow$ Odd terms vanish

$j=0:$ $a_2 = \frac{1\cdot 2 - m(m+1)}{2\cdot 3} a_0 = -\frac{(m-1)(m+2)}{3!} a_0$

$j=2:$ $a_4 = \frac{3\cdot 4 - m(m+1)}{4\cdot 5} a_2 = \frac{12 - m(m+1)}{20} a_2$

$$= \frac{(m-3)(m-1)(m+2)(m+4)}{5!} a_0$$

$$y_2(x) = a_0\left[x - \frac{(m-1)(m+2)}{3!}x^3 + \frac{(m-3)(m-1)(m+2)(m+4)}{5!}x^5 \dots\right]$$

Q12 -

$$e^{i\vec{k}\cdot\vec{r}} = e^{ikr\cos\theta}$$

$$\cancel{= \sum_{l=0}^{\infty} a_l P_l(x)}$$

$$= \sum_{l=0}^{\infty} C_l P_l(\cos\theta)\, j_l(kr)$$

Differentiating w.r.t. kr

$$i\cos\theta\, e^{ikr\cos\theta} = \sum_{l=0}^{\infty} C_l\, j_l'(kr)\, P_l(\cos\theta)$$

$$i\cos\theta \sum_{l=0}^{\infty} C_l\, P_l(\cos\theta)\, j_l(kr) = \sum_{l=0}^{\infty} C_l\, P_l(\cos\theta)\, j_l'(kr)$$

$$\cos\theta\, P_l(\cos\theta) = \frac{l+1}{2l+1} P_{l+1}(\cos\theta) + \frac{l}{2l+1} P_{l-1}(\cos\theta)$$

$$\Rightarrow i\left[\frac{m}{2m-1} C_{m-1}\, j_{m-1}(kr) + \frac{m+1}{2m+3} C_{m+1}\, j_{m+1}(kr)\right]$$

$$= \frac{C_m}{2m+1}\left[m\, j_{m-1}(kr) - (m+1)\, j_{m+1}(kr)\right]$$

Equating coeffs. of j_{m-1}:

$$C_m = i\,\frac{(2m+1)}{(2m-1)}\, C_{m-1}$$

$$\left.\begin{aligned} m=1 &: C_1 = i\cdot 3\, C_0 \\ m=2 &: C_2 = i^2\cdot 5\cdot C_0^2 \\ m=3 &: C_3 = i^3\cdot 7\cdot C_0^3 \end{aligned}\right\} \quad C_m = i^m (2m+1)\, C_0^m$$

$$e^{ikr\cos\theta} = C_0\, j_0(kr)\, P_0(\cos\theta) + C_1\, j_1(kr)\, P_1(\cos\theta) + \dots$$

$kr \to 0$

$$1 = C_0 + 0 + 0 + \dots$$

$$C_0 = 1$$

$$\therefore C_m = i^m (2m+1)$$

$$\therefore e^{ikr\cos\theta} = \sum_{l=0}^{\infty} i^l (2l+1)\, j_l(kr)\, P_l(\cos\theta)$$

$$\int_V e^{i\vec{k}\cdot\vec{r}}\, d^3r = \int_V e^{ikr\cos\theta}\, r^2 \sin\theta\, dr\, d\theta\, d\phi$$

$$= \int_V \sum_{l=0}^{\infty} i^l (2l+1) j_l(Kr)\, P_l(\cos\theta) \times P_0(\cos\theta) \times r^2 \sin\theta\, dr\, d\theta\, d\phi$$

$$= 2\pi \sum_{l=0}^{\infty} \int_0^R dr\, r^2\, i^l (2l+1) j_l(Kr) \underbrace{\int_0^{\pi} P_l(\cos\theta) P_0(\cos\theta) \sin\theta\, d\theta}_{2\delta_{l,0}}$$

$$= 2\pi \times 2 \int_0^R dr\, j_0(Kr)\, r^2$$

$$t = Kr, \quad dt = K\,dr$$

$$= 4\pi \int_0^{KR} \frac{dt}{K}\, j_0(t) \cdot \frac{t^2}{K^2}$$

$$= \frac{4\pi}{K^3} \int_0^{KR} t^2 j_0(t)\, dt = \frac{4\pi}{K^3} \int_0^{KR} \frac{d}{dt}\left[t^2 j_1(t)\right] dt$$

$$= \frac{4\pi}{\cancel{k^3}} \times \cancel{k^2} R^2 j_1(kR)$$

$$= \frac{4\pi R^2}{k} j_1(kR)$$

$$\int_V e^{i\vec{k}\cdot\vec{r}}\, d^3r = 4\pi R^3 \frac{j_1(kR)}{kR}$$